Taekhyun Park

Busan, South Korea

Email | GitHub | Homepage
Google Scholar | LinkedIn | Hugging Face

Research Interests

- **Novel architectures:** JEPA, Deep Supervision, State Space Models, and Looped LLMs
- **Sequence modeling:** Time Series, Language, Video, Streaming Data Based World Modeling
- **LLM systems:** Process Mining with LLMs, RLVR, and Agent-Based Systems

Education

Pusan National University Mar. 2025 – Feb. 2027
M.S. in Data Science, Graduate School of Data Science Busan, South Korea
– Advisor: Prof. Hyerim Bae

Korea Maritime & Ocean University Mar. 2019 – Feb. 2025
B.S. in Control Engineering Busan, South Korea
– Completed military service in the Republic of Korea Army, Oct. 2020 – Apr. 2022

Projects

AI기반 빅데이터 통계시스템 구축 및 체인포털 서비스 개선 사업 June. 2026 – Dec. 2026
Development of an AI-Based Big Data Statistics System and Enhancement of Chain Portal Services
Funded by Busan Port Authority (BPA)

해상교통 환경 특화 대형행동모형 개발을 위한 에이전트 기술 개발 June. 2026 – Dec. 2026
Development of Agent Technologies for a Large Action Model Specialized in Maritime Traffic Environments
Funded by Korea Research Institute of Ships & Ocean Engineering (KRISO)

프로세스 마이닝을 활용한 고객 사용 패턴 기반 개인화 스마트 운전 로직 개발 Dec. 2025 – Nov. 2026
Development of Personalized Smart Operation Logic Based on Customer Usage Patterns Using Process Mining
Funded by LG Electronics

이상치 탐지 배원시뮬레이터 개발 Oct. 2025 – Dec. 2025
Development of a Workforce Allocation Simulator with Anomaly Detection
Funded by Samsung Heavy Industries

동적 해양사고 시나리오 도출 및 실시간 사고위험 통합지표 기반 조기경보 시스템 개발 Aug. 2025 – Jan. 2026
Development of Dynamic Maritime Accident Scenarios and an Early Warning System Based on an Integrated Real-Time Accident Risk Index
Funded by Pusan National University and RISE

인간 중심 - 탄소 중립 글로벌 공급망 연구센터 Mar. 2025 – Feb. 2027
Research Center for Human-Centered, Carbon-Neutral Global Supply Chains
Funded by the National Research Foundation of Korea

Publications

Journals & Conference Proceedings

[J] Generative AI and Machine learning collaboration for container dwell time prediction via data standardization TRE, 2026

– Minseop Kim, Takhyeong Kim, **Taekhyun Park**, Hanbyeol Park, and Hyerim Bae
– *Transportation Research Part E: Logistics and Transportation Review*, Vol. 216, Article 105171.
– DOI: 10.1016/j.tre.2026.105171

[I] Design and Implementation of an LLM-Based Maritime Risk Scenario Generation System for Vessel Traffic Services IEEE CASE, 2026

– **Taekhyun Park***, Sangmin Jo, Dohee Kim, and Hyerim Bae

[J] LURKER: Language Models for Unveiling Maritime Risks via Knowledge-Driven Event Forecasting and Reasoning ICIC Express Letters

– Seongmoon Hong*, **Taekhyun Park**, Dohee Kim, Sangmin Jo, and Hyerim Bae

[J] Identifying Severity of Maritime Accidents Based on Weighted Kernel Density Estimation ICIC Express Letters

– Sangmin Jo*, Dohee Kim, Seongmoon Hong, **Taekhyun Park**, and Hyerim Bae

[I] JustDense: Just using Dense instead of Sequence Mixer for Time Series Analysis IEEE BigData 2025

– **Taekhyun Park***, Yongjae Lee*, Daesan Park*, Dohee Kim, and Hyerim Bae
– Code

[J] LA-RCS: LLM-agent-based Robot Control System Sensors and Materials

– **Taekhyun Park***, Young-Jun Choi, Seung-Hoon Shin, Chang-Eun Lee, and Kwangil Lee
– Project Page

Preprints

Variable-Horizon Workforce Demand Forecasting with an Aggregate Demand Constraint for Construction Workforce Planning arXiv, 2026

– Hanbyeol Park*, Jaehyeon Heo*, **Taekhyun Park**, Minseong Kim, and Hyerim Bae

Trie-Constrained Token Prediction with Hierarchy-Aware Semantic Alignment for HS Code Prediction arXiv, 2026

– Minseop Kim*, **Taekhyun Park**, Kikun Park, and Hyerim Bae

LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models arXiv, 2026

– **Taekhyun Park***, Yongjae Lee, Dohee Kim, and Hyerim Bae
– Project Page

Rewarding Structural Conformance of Reasoning using Process Mining arXiv, 2025

– Yongjae Lee*, **Taekhyun Park***, and Hyerim Bae
– Code

Application of Large Language Models for Container Throughput Forecasting: Incorporating Contextual Information in Port Logistics arXiv, 2026

– Minseop Kim*, Jaeeun Kwon, Hanbyeol Park, Kikun Park, **Taekhyun Park**, and Hyerim Bae

Selected Presentations

Online Learning — Weekly Lab Seminar, Sep. 2026
Qwen: Global AI Strategy and Technical Ecosystem — SCSC LLM WG Seminar, Jul. 2026
Introduction of JEPA — Weekly Lab Seminar, Jun. 2026
Recursive Model — Weekly Lab Seminar, Feb. 2026
BitNet Overview — Freshman Study, Feb. 2025
From RNN to Mamba2: Solving Online Function Approximation Problems in Post-LLM Models — Freshman Study, Feb. 2025

Academic Service

The 20th ACM Conference on Web Search and Data Mining (WSDM 2027) — Reviewer

Internship

DeepLogicChain — Jan. 2024 – Mar. 2024
Intern — Seoul, South Korea

Teaching Experience

Pusan National University — Mar. 2025 – Jun. 2026
Teaching Assistant for Data Structures and Algorithms (Undergraduate Course) — Busan, South Korea

Honors & Awards

부산테크노파크 원장상 — 2025
K-해양 AI 챌린지 / K-Maritime AI Challenge
실시간 데이터 및 해양 규제 융합형 LLM 기반 사고 대응 지원 시스템
LLM-Based Incident Response Support System Integrating Real-Time Data and Maritime Regulations

레인보우브레인상 — 2023
KMOU 캡스톤디자인 / KMOU Capstone Design
RAG 및 피드백 기반 AI 기법을 활용한 학교 게시판 검색 AI 시스템
AI Search System for University Bulletin Boards Using RAG and Feedback-Based AI Techniques

대상 — 2022
KMOU AI 아이디어 챌린지 / KMOU AI Idea Challenge
이미지 생성 AI 학습을 위한 P2P 기반 데이터셋 구축 및 AI 소유권 가상자산화 플랫폼
P2P Platform for Building Image-Generation AI Training Datasets and Tokenizing AI Ownership

경남테크노파크 원장상 — 2022
AI & 블록체인 아이디어 경진대회 / AI & Blockchain Idea Competition
블록체인 기술을 활용한 글쓰기 하이퍼 플랫폼
Blockchain-Based Writing Hyperplatform

Technical Skills

Languages: Python, R, Go, C/C++, Ruby, VHDL, MATLAB, Solidity, Elixir
Tools: CUDA, Triton, PyTorch